

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**  
**Work Integrated Learning Programmes Division**  
**First Semester 2024-2025**  
**Mid-Semester Test**  
**(EC-2 Regular)**

Course No. : AIMLCZC418  
Course Title : Introduction to Statistical Methods  
Nature of Exam : Closed Book  
Weightage : 30%  
Duration : 2 Hours  
Date of Exam : 18.01.2025 FN

|                  |     |
|------------------|-----|
| No. of Pages     | = 4 |
| No. of Questions | = 6 |

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

**Answer all the questions:**

Q1. a) Consider waiting times (in minutes):

4.21, 5.55, 3.02, 5.13, 4.77, 2.34, 3.54, 3.20, 4.50, 6.10, 0.38, 5.12, 6.46, 6.19, 3.79.

Determine the five-number summary: Minimum, Q1, Median, Q3, Maximum. Based on the results obtained, give an interpretation about the data. [3Marks]

Sorted Data: 0.38, 2.34, 3.02, 3.20, 3.54, 3.79, 4.21, 4.50, 4.77, 5.12, 5.13, 5.55, 6.10, 6.19, 6.46.

Five-number Summary: Min = 0.38, Q1 = 3.20, Median = 4.50, Q3 = 5.55, Max = 6.46.

Interpretation: The data is slightly positively skewed due to the longer tail on the higher values.

b) The SpaceX space program lasted from 2020 until 2024 and included 13 launches. The missions lasted from as little as 7 hours to as long as 301 hours. The duration of each launch is listed below: 9, 195, 241, 301, 216, 260, 7, 244, 192, 147, 10, 295, 142. Find the mean, standard deviation of the launch times. [2Marks]

|                                        |               |
|----------------------------------------|---------------|
| <b>Mean = 173.769231</b>               | <b>1 mark</b> |
| <b>Standard deviation = 105.611832</b> | <b>1 mark</b> |

Q2. A study on ethics in the workplace by the Ethics Resource Center and Kronos, Inc., revealed that 35% of employees admit to keeping quiet when they see co-worker misconduct. Suppose 75% of employees who admit to keeping quiet when they see co-worker misconduct call in sick when they are well. In addition, suppose that 40% of the employees who call in sick when they are well admit to keeping quiet when they see co-worker misconduct. If an employee is randomly selected, determine the following probabilities:

- a. The employee calls in sick when well and admits to keeping quiet when seeing co-worker misconduct.
  - b. The employee admits to keeping quiet when seeing co-worker misconduct or calls in sick when well.
  - c. Given that the employee calls in sick when well, he or she does not keep quiet when seeing co-worker misconduct.
  - d. The employee neither keeps quiet when seeing co-worker misconduct nor calls in sick when well.
  - e. The employee admits to keeping quiet when seeing co-worker misconduct and does not call in sick when well
- [5Marks]

Sol:

5 Let Q = keep quiet when they see co-worker misconduct

Let C = call in sick when they are well

$$P(Q) = .35 \quad P(NQ) = 1 - .35 = .65 \quad P(C|Q) = .75 \quad P(Q|C) = .40$$

$$a) P(C \cap Q) = P(Q) \cdot P(C|Q) = (.35)(.75) = \mathbf{.2625}$$

$$b) P(Q \cup C) = P(Q) + P(C) - P(C \cap Q)$$

but  $P(C)$  must be solved for:

$$P(C \cap Q) = P(C) \cdot P(Q|C)$$

$$.2625 = P(C) (.40)$$

$$\text{Therefore, } P(C) = .2625 / .40 = .65625$$

$$\text{and } P(Q \cup C) = .35 + .65625 - .2625 = \mathbf{.74375}$$

$$c) P(NQ|C) = P(NQ \cap C) / P(C)$$

$$\text{but } P(NQ \cap C) = P(C) - P(C \cap Q) = .65625 - .2625 = .39375$$

$$\text{Therefore, } P(NQ|C) = P(NQ \cap C) / P(C) = .39375 / .65625 = \mathbf{.60}$$

$$d) P(NQ \cap NC) = 1 - P(Q \cup C) = 1 - .74375 = \mathbf{.25625}$$

$$e) P(Q \cap NC) = P(Q) - P(Q \cap C) = .35 - .2625 = \mathbf{.0875}$$

Q 3. a) Based on past experience, it is assumed that the number of flaws per foot in rolls of grade 2 paper follows a Poisson distribution with a mean of 0.2 flaw per foot. What is the probability that in a

i. 1-foot roll, there will be at least 2 flaws?

ii. 50-foot roll, there will be greater than or equal to 5 flaws and less than or equal to 8 flaws

[3Marks]

Sol:

a) ANSWER:  $\lambda = 0.2$

Using the formula  $P(X \geq k) = 1 - P(X < k)$ , where X follows a Poisson distribution, we can calculate

$$i) P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$$

$$= 1 - [e^{(-0.2)}(0.2)^0/0! + e^{(-0.2)}(0.2)^1/1!] = 1 - 0.9824$$

$$= 0.0176$$

The **probability** of having at least 2 flaws in a 1-foot roll of grade 2 paper is 0.0176.

The problem describes a Poisson distribution with a mean of  $\lambda = 0.2$  flaws per foot. For a 50-foot roll, the mean number of flaws is:

$$\lambda_{\text{roll}} = 50 \times 0.2 = 10$$

The probability of observing  $k$  flaws in a Poisson distribution is given by:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where:

- $\lambda = 10$
- $k$  is the number of flaws.

We want the probability of having  $5 \leq X \leq 8$ . This is:

$$P(5 \leq X \leq 8) = P(X = 5) + P(X = 6) + P(X = 7) + P(X = 8)$$

Each term can be computed as:

$$P(X = k) = \frac{10^k e^{-10}}{k!}$$

I'll calculate this probability step by step.

The probabilities for each number of flaws are:

$$P(X = 5) = 0.0378, P(X = 6) = 0.0631, P(X = 7) = 0.0901, P(X = 8) = 0.1126$$

The total probability for  $5 \leq X \leq 8$  is:

$$0.3036$$

b) An individual has 3 different email accounts. Most of her messages, in fact 70%, come into account #1, whereas 20% come into account #2 and the remaining 10% into account #3. Of the messages into account #1, only 1% are spam, whereas the corresponding percentages for accounts #2 and #3 are 2% and 5%, respectively.

What is the probability that a randomly selected message is spam? [2M]

Sol:

Let S be the event that a message is spam.

Let A1 be the event that the message arrives in account #1.

Let A2 be the event that the message arrives in account #2.

Let A3 be the event that the message arrives in account #3.

List Given Probabilities

$$P(A1) = 0.70, P(A2) = 0.20, P(A3) = 0.10$$

$$P(S|A1) = 0.01, P(S|A2) = 0.02, P(S|A3) = 0.05$$

Calculate the Probability of a Spam Message

We can use the Law of Total Probability to find P(S):

$$P(S) = P(S|A1) * P(A1) + P(S|A2) * P(A2) + P(S|A3) * P(A3)$$

$$P(S) = (0.01 * 0.70) + (0.02 * 0.20) + (0.05 * 0.10)$$

$$P(S) = 0.007 + 0.004 + 0.005$$

$$P(S) = 0.016$$

Therefore, the probability that a randomly selected message is spam is 0.016 or 1.6%.

Q 4. a) The discrete random variables  $X$  and  $Y$  have joint probability mass function [3Marks]

$$f(x, y) = \begin{cases} cxy & x = 1, 2, 3, y = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

- Find the value of constant  $c$ .
- Find the marginal probability function for  $X$  and  $Y$ .

- b) From a lot of 10 items containing 3 defective items, a sample of 4 items is drawn at random. Let the random variable  $X$  denote the number of defective items in the sample. Answer the following when the sample is drawn without replacement [2Marks]
- Find the probability distribution of  $X$ .
  - Find  $P(X \leq 1)$
  - Find  $P(0 < X \leq 2)$

Sol:

a). i)

$$\therefore \sum_x \sum_y cxy = 1$$

$$\therefore c + 2c + 2c + 4c + 3c + 6c = 1$$

$$\therefore 18c = 1$$

$$\therefore c = \frac{1}{18}$$

ii) Let  $g(x)$  be marginal probability function of  $x$ .

$$\therefore g(x) = \sum_y f(x, y) = \sum_y cxy = cx(1 + 2) = \frac{3x}{18} = \frac{x}{6}, x = 1, 2, 3.$$

Let  $h(y)$  be marginal probability function for  $y$ :

$$h(y) = \sum_x f(x, y) = \sum_x cxy = cy(1 + 2 + 3) = 6cy = \frac{6y}{18} = \frac{y}{3}, y = 1, 2.$$

b)

i) Let  $X$  be random variable with possible values are **0, 1, 2, 3**. We calculate the probability distribution for each by using following formula:

$$P(X = k) = \frac{{}^3C_k {}^7C_{4-k}}{{}^{10}C_4}.$$

Therefore, probability mass function for  $X$  is:

|      |     |     |      |      |
|------|-----|-----|------|------|
| X    | 0   | 1   | 2    | 3    |
| P(X) | 1/6 | 1/2 | 3/10 | 1/30 |

$$ii) P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{1}{6} + \frac{1}{2} = \frac{2}{3}.$$

$$iii) P(0 < x \leq 2) = P(1) + P(2) = \frac{1}{2} + \frac{3}{10} = \frac{4}{5}.$$

[1M]

[0.5M]

[0.5M]

Q 5. If the weather is Snowy, then the player will play or not?

[5Marks]

| Weather | Player play | Weather | Player play |
|---------|-------------|---------|-------------|
| Sunny   | Yes         | Cloudy  | Yes         |
| Rainy   | No          | Sunny   | Yes         |
| Cloudy  | Yes         | Snowy   | No          |
| Sunny   | No          | Cloudy  | Yes         |
| Sunny   | Yes         | Rainy   | No          |
| Snowy   | No          | Snowy   | No          |
| Rainy   | No          | Snowy   | Yes         |

Sol:

We need to calculate

$P(\text{Player Play} = \text{Yes} \mid \text{Weather} = \text{Snowy})$ .

Bayes' theorem states:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Here: A: Player Play = Yes, B: Weather = Snowy

**Step 1: Compute  $P(\text{Player Play} = \text{Yes})$**

Count the rows where "Player Play = Yes" and divide by the total number of rows.

Total rows: 14

Rows with "Player Play = Yes": 7

$$P(\text{Player Play} = \text{Yes}) = \frac{7}{14} = 0.5$$

**Step 2: Compute  $P(\text{Weather} = \text{Snowy} \mid \text{Player Play} = \text{Yes})$**

Count the rows where:

"Weather = Snowy"

"Player Play = Yes".

From the table:

Rows with "Snowy" and "Yes": 1 (Row 14)

Rows with "Player Play = Yes": 7

$$P(\text{Weather} = \text{Snowy} \mid \text{Player Play} = \text{Yes}) = \frac{1}{7} \approx 0.1429$$

**Step 3: Compute  $P(\text{Weather} = \text{Snowy})$**

Count all rows where "Weather = Snowy" and divide by the total number of rows.

Rows with "Weather = Snowy": 4 (Rows: 6, 12, 13, 14)

$$P(\text{Weather} = \text{Snowy}) = \frac{4}{14} \approx 0.2857$$

**Final Calculation**

Using Bayes' theorem:

$$= \frac{P(\text{Weather} = \text{Snowy} \mid \text{Player Play} = \text{Yes}) \cdot P(\text{Player Play} = \text{Yes})}{P(\text{Weather} = \text{Snowy})}$$

Substitute the values:

$$P(\text{Player Play} = \text{Yes} \mid \text{Weather} = \text{Snowy}) = \frac{0.1429 \cdot 0.5}{0.2857} \approx 0.25$$

The probability that the player will play when the weather is snowy is 0.25 (25%). Based on this, it is less likely that the player will play in snowy weather.

Q 6. a) Incidence of a rare disease. Only 1 in 1000 adults is afflicted with a rare disease for which a diagnostic test has been developed. The test is such that when an individual actually has the disease, a positive result will occur 99% of the time, whereas an individual without the disease will show a positive test result only 2% of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease? [3Marks]

b) A certain machine makes electrical resistors having mean resistance of 40 ohms and standard deviation of 2 ohms. If the resistance follows a normal distribution and can be measured to any degree of accuracy.

- i) What percentage of resistors have resistances that exceed 43 ohms?
- ii) What percentage of resistors will have resistances that exceed 43 ohms to the nearest ohm? [2Marks]

Sol:

Let D be the event that the individual has the disease.

Let D' be the event that the individual does not have the disease.

Let + be the event that the test result is positive.

Let - be the event that the test result is negative.

List Given Probabilities

$$P(D) = 1/1000 = 0.001 \text{ (Prevalence of the disease)}$$

$$P(D') = 1 - P(D) = 0.999$$

$$P(+|D) = 0.99 \text{ (Sensitivity of the test)}$$

$$P(+|D') = 0.02 \text{ (False positive rate)}$$

Calculate the Probability of a Positive Test Result

We can use the Law of Total Probability to find P(+):

$$P(+) = P(+|D) * P(D) + P(+|D') * P(D')$$

$$P(+) = (0.99 * 0.001) + (0.02 * 0.999)$$

$$P(+) = 0.00099 + 0.01998$$

$$P(+) = 0.02097$$

Calculate the Probability of Having the Disease Given a Positive Test Result (Using Bayes' Theorem)

$$P(D|+) = (P(+|D) * P(D)) / P(+)$$

$$P(D|+) = (0.00099) / 0.02097$$

$$P(D|+) \approx 0.0472$$

Therefore, the probability that the individual has the disease given a positive test result is approximately 0.0472 or 4.72%.

b)

Given  $\mu = 40, \sigma = 2$

1. To find

$$\begin{aligned} P(X > 43) &= P\left(\frac{X - 40}{2} > \frac{43 - 40}{2}\right) \\ &= P(Z > 1.5) = 0.5 - P(0 < Z < 1.5) \\ &= 0.5 - 0.4332 \\ &= 0.0668 \end{aligned}$$

$\therefore$  6.68% of resistors will have resistance exceeding 43 ohms.

2. Nearest to the 43 ohm means between 42.5 and 43.5. So

$$\begin{aligned} P(X > 43.5) &= P\left(\frac{X - 40}{2} > \frac{43.5 - 40}{2}\right) \\ &= P(Z > 1.75) = 0.5 - P(0 < Z < 1.75) \\ &= 0.5 - 0.4599 = 0.0401 \end{aligned}$$

$\therefore$  4.01% of resistors will have resistances exceeding 43 ohms to the nearest ohm.